

Requirement Analysis

Technology Stack (Architecture & Stack)

Date	09 September 2026
Student Name	Dwarapudi Nagasri
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	

Technical Architecture

Technology Stack & Layered Architecture

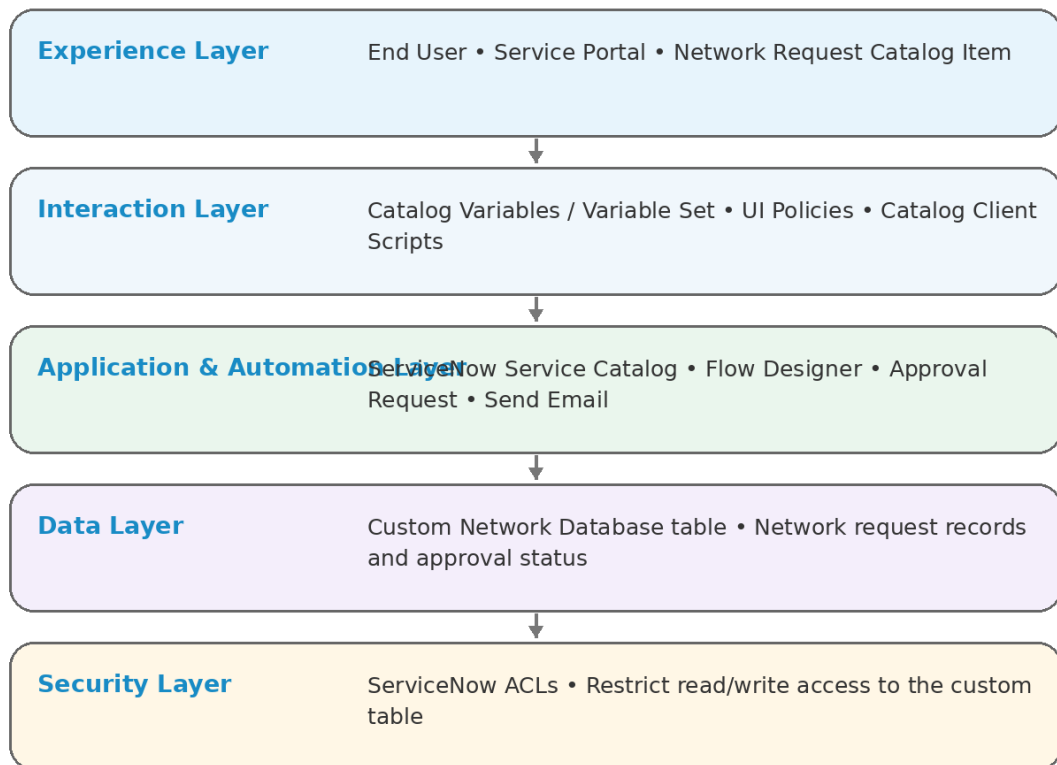


Figure 1: Layered technology architecture for the Automated Network Request Management solution.

The solution is implemented within ServiceNow using the Service Portal, Service Catalog, catalog variables and variable sets, Flow Designer, a custom Network Database table, approval requests, UI Policies, catalog client scripts, and ACLs. No external API, machine-learning model, or separate cloud database is specified in the supplied project instructions.

Figure 1: ServiceNow technology stack and request-management architecture flow.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Service Portal through which end users search for Network Requests, fill required details, and submit requests.	ServiceNow Service Portal
2	Application Logic-1	Catalog request handling and variable configuration.	ServiceNow Service Catalog
3	Application Logic-2	End-to-end request lifecycle automation including Get Catalog Variables, Create Record, Send Email, Ask for Approval, If/Else, and Update Record.	ServiceNow Flow Designer
4	Application Logic-3	Dynamic field visibility and dependent form behaviour.	Catalog UI Policies and Catalog Client Scripts
5	Database	Stores network request records and fields including Database Number, Assignment To, Mobile Number, Total Amount, Mode of Payment, Type of Connection, Customer Address, Approval Status, and Requested For.	ServiceNow custom Network Database table
6	Approval	Stores/displays approval request information related to the Network Database record.	ServiceNow Approval Request / sysapprovals relationship
7	Email Notification	Communicates request progress from the automated flow.	ServiceNow Flow Designer Send Email action
8	External API-1	No external API is specified in the project instructions.	Not specified
9	Machine Learning Model	No machine-learning model is specified in the project instructions.	Not specified
10	Infrastructure (Server / Cloud)	Development and testing are performed in a ServiceNow Personal Developer Instance (PDI).	ServiceNow PDI

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No external open-source framework is specified.	Not specified
2	Security Implementations	Default ACLs on the custom table are used to restrict read/write access according to configured access controls.	ServiceNow ACLs
3	Scalable Architecture	The solution uses a structured custom table and automated Flow Designer process; no specific microservices or scaling technology is specified.	ServiceNow platform
4	Availability	The solution runs within the ServiceNow platform/PDI; no separate availability infrastructure is specified.	ServiceNow platform
5	Performance	Project objectives include reduced manual effort and faster request fulfilment; no requests-per-second, caching, or CDN metric is specified.	ServiceNow Flow Designer / Service Portal

